Mohak Kapoor

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SUMMARY

ML Engineer with 1+ year of experience building and deploying production machine learning systems on edge hardware and cloud infrastructure. Proficient in Python, XGBoost, PyTorch, TensorFlow, FastAPI, Docker, and SQL. Experienced in deep learning, time-series forecasting, anomaly detection, LLM integration, and MLOps pipeline development. Seeking ML Engineer, AI Engineer, or Data Scientist roles.

TECHNICAL SKILLS

Languages	Python (Advanced), TypeScript, C++
ML & Modeling	Supervised Learning, SVMs, LightGBM, XGBoost, CNN, BiLSTM, CRNN, Time-Series Forecasting, Anomaly Detection, Statistical Modeling, Feature Engineering, Cross-Validation, Hyperparameter Tuning
LLMs & GenAI	LLM Integration (GPT-4, Gemini), Prompt Engineering, Multimodal Pipelines, MCP, RAG
MLOps & Deployment	FastAPI, REST APIs, Docker, CI/CD (GitHub Actions), MLflow, Experiment Tracking, Model Monitoring, Cloudflare Workers, Cloudflare Tunnel, Edge Deployment (Raspberry Pi 5)
Frameworks & Data	TensorFlow, PyTorch, scikit-learn, Django, Hugging Face, Pandas, NumPy, Polars, PostgreSQL, SQL, Jupyter, Git, Linux
CS Fundamentals	Data Structures & Algorithms, Object-Oriented Programming, Computer Networks

EXPERIENCE

HumanizeIQ — *June 2025 – October 2025*
AI Intern — *Hazlet, New Jersey (Remote)*

- Architected and deployed **Model Context Protocol (MCP) servers** on **serverless** Cloudflare Workers for asynchronous image generation and multimodal document workflows, with persistent object storage via Cloudflare R2.
- Designed recruiter call analysis pipelines integrating **LLMs (GPT-4, Gemini)** for **NLP-driven** automated transcription analysis and structured report generation, **eliminating 100% of manual post-call processing** and reducing analyst turnaround to zero.
- Built **REST APIs** for background job orchestration and **scalable asynchronous processing**, enabling reliable long-running inference tasks within chat-based LLM workflows.

PROJECTS

METR — Market Exposure Timing Research — Python, Polars, XGBoost, scikit-learn, Feature Engineering — *Apr 2026*

- Constructed a **two-layer meta-labeling framework** on 3,440+ days of multi-asset Indian market data (Nifty 50, GoldBees, USD/INR), using Polars with signal-conditional masking and 30 cross-asset features for **time-series classification** and trade entry quality filtering via XGBoost.
- Validated against **10,000 Monte Carlo simulations** — GoldBees filter achieved **Sharpe 1.48, Calmar 1.87, 66.1% win rate OOS (2024–2025)**, significant at $p<0.05$ across three independent tests; null results on Nifty and USD/INR consistent with market efficiency theory.
- Applied **Fractional Differentiation** (d=0.45) preserving 81% historical memory with stationarity, **Triple Barrier volatility-adaptive labeling**, and **SHAP feature importance** analysis — identifying Vol Efficiency and VIX Momentum as primary alpha drivers consistent with GoldBees' rupee-dollar exposure.

Network IDS Detection Engine — Python, LightGBM, XGBoost, PyTorch, scikit-learn, FastAPI, cuML — *Sep 2025 – Present*

- Engineered a **dual-detection intrusion detection system** (signature-based + unsupervised anomaly detection) on 2.8M rows of CICIDS2017 network traffic, mirroring production platforms like Snort and Darktrace — temporal train/test split and **imbalanced class handling** to prevent distribution leakage across unseen attack types.
- Benchmarked LightGBM at **99% accuracy (F1: 0.99)**, XGBoost 97%, PyTorch FFNN 98%, SVM 96.7% — IncrementalPCA **reduced feature space 51%** (69 to 34 features, 98.97% variance retained). Denoising Autoencoder (AUC: 0.7801) outperforms Isolation Forest (AUC: 0.7156) on zero-day threat detection.
- Shipped inference engine as a **FastAPI service on Raspberry Pi 5** (no GPU) with **model serving** architecture, exposed publicly via **Cloudflare Tunnel** — modular REST backend with dedicated prediction, routing, and CORS layers.

CaptchaOCR — CAPTCHA Recognition System — Python, PyTorch, CNN, BiLSTM, CTC Loss, Hugging Face — *Dec 2025*

- Trained a CAPTCHA recognition system using a **CNN + BiLSTM (CRNN) architecture with CTC loss**, achieving **96%+ character-level accuracy** on distorted, variable-length sequences via synthetic training corpus generation and noise-augmentation preprocessing.
- Published to **Hugging Face Spaces** for public **model deployment** and inference, enabling reproducible benchmarking and open community validation beyond offline experimentation.

EDUCATION

Jaypee Institute of Information Technology — *Noida, India*
B.Tech in Electronics and Communication Engineering — *Sept 2022 – July 2026*

- **Relevant Coursework:** Machine Learning (A+), Deep Learning (A+), Advanced Statistics (A), Computer Systems, Multivariate & Differential Calculus, Control Systems